

This project aims to calculate key performance indicators for a fictional company called Gamezone. The data is initially stored in Excel, and any irregularities are identified and evaluated. Solvable Discrepancies are removed using Excel Power Query, and insurmountable discrepancies are ignored. Following this, the data is used to create a dashboard in Power BI.

Each row represents an order, and the identifier is the order ID.

The key metric is the Price in USD.

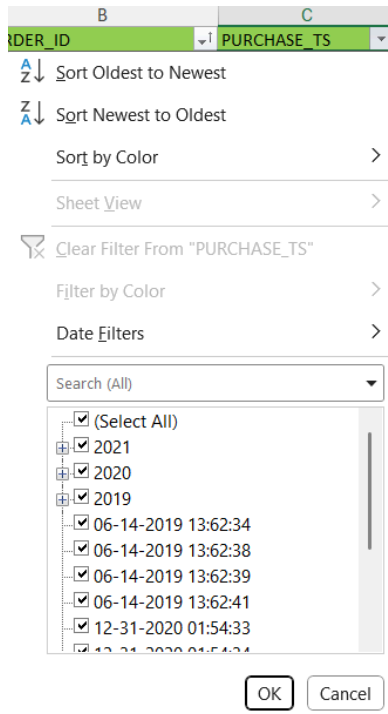
The key dimensions are the time, location of purchase, shipping date, the product name, marketing channels, and the country code.

As an added example, if we wanted to identify the countries with the most sales, our key metric would be the geographical data.

A small snippet of the data:

	A	B	C	D	E	F	G	H	I	J	K
1	USER_ID	ORDER_ID	PURCHASE_TS	SHIP_TS	PRODUCT_NAME	PRODUCT_ID	USD_PRICE	PURCHASE_PLATFORM	MARKETING_CHANNEL	ACCOUNT_CRE	COUNTRY_CODE
2	2c06175e	0001328c3c220830	2020-12-24	2020-12-13	Nintendo Switch	e682	168	website	affiliate	unknown	US
3	ee8e5bc2	0002af7a5c6100772	2020-10-01	2020-09-21	Nintendo Switch	e682	160.61	website	direct	desktop	DE
4	9eb4efe0	0002b8350e167074	2020-04-21	2020-02-16	Nintendo Switch	8d0d	151.2	website	direct	desktop	US
5	cac7cbaf	0006d06e98385729	2020-04-07	2020-04-04	Sony PlayStation 5 Bundle	54ed	1132.82	website	direct	desktop	AU
6	6b0230bc	00097279a2f46150	2020-11-24	2020-08-02	Nintendo Switch	8d0d	33.89	website	direct	desktop	TR
7	04e67528	000f169e17f4941	2020-08-20	2020-08-09	Nintendo Switch	8d0d	168	website	direct	desktop	PH
8	94dfec76	0006d99f6c3101	2020-05-11	2020-03-05	27in 4K gaming monitor	891b	390.59	website	direct	desktop	GB
9	8a328a1d	0010d31e3c058676	2020-07-03	2020-04-03	27in 4K gaming monitor	891b	480	website	direct	desktop	BR
10	46c57b58	0013ac51a8831707	2019-03-24	2018-10-29	JBL Quantum 100 Gaming Headset	ab0f	22.22	website	direct	desktop	BE
11	a2eaa1d1	001620ea42468912	2019-11-01	2019-10-17	Nintendo Switch	e682	168	website	direct	desktop	US
12	c7c492f8	0016654fc8584512	2019-12-09	2019-07-22	Nintendo Switch	8d0d	150.53	website	direct	desktop	FR
13	d71c06c6	00236994a0b90874	2019-08-30	2019-06-29	Nintendo Switch	8d0d	146.25	website	direct	desktop	KR
14	9933b602	002a0b04dd764807	2020-02-16	2019-12-08	Sony PlayStation 5 Bundle	54ed	1725.86	website	direct	desktop	US
15	bbbaa7ff	002f0b22bb379479	2020-05-14	2020-03-24	Nintendo Switch	e682	168	website	direct	desktop	US
16	68673786	003ed9bd3d71799	2020-10-26	2020-06-01	Nintendo Switch	8d0d	134.41	website	direct	desktop	ES
17	1802f1a0	004602d9c2d12771	2020-10-13	2020-08-30	Sony PlayStation 5 Bundle	54ed	1368.45	website	direct	desktop	AU
18	2abdb1d2	004d6bc82f820293	2020-10-25	2020-08-01	27in 4K gaming monitor	e7e6	480	website	direct	desktop	US
19	deba2f70	004eca8109c94135	2019-12-24	2019-09-11	Nintendo Switch	e682	168	website	direct	desktop	US
20	d9f8bd41	0053e1f565292085	2019-05-20	2019-04-09	JBL Quantum 100 Gaming Headset	ab0f	24	website	direct	desktop	US
21	d37ef9f9	006116b394689293	2020-12-23	2020-08-29	27in 4K gaming monitor	891b	312	website	direct	desktop	US
22	1962f4de	00617f69ac813297	2019-03-23	2019-03-08	JBL Quantum 100 Gaming Headset	ab0f	19.2	website	direct	desktop	US
23	6ee318a8	0064a61f5e347824	2020-12-06	2020-08-30	JBL Quantum 100 Gaming Headset		2997	24 mobile app	direct	desktop	US

Issues are then identified. Here is an example of date discrepancies in the purchase_ts column:



We can see that there are dates that are formatted incorrectly, and that there is an inconsistency in the dates. We can filter the data to find these inconsistencies.

	USER_ID	ORDER_ID	PURCHASE_TS	SHIP_TS	PRODUCT_NAME	PRODUCT_ID	USD_PRICE	PURCHASE_PLATFORM	MARKETING_CHANNEL	ACCOUNT_CRE	COUNTRY_CODE
1049	a5298a4d	0dda212aae69940		2019-07-08	JBL Quantum 100 Gaming Headset	ab0f	21.96	website	direct	desktop	FR
1253	e749986a	1041360a9a997694	12-31-2020 01:54:35	2020-11-03	27in 4K gaming monitor	891b	480	website	direct	desktop	US
5848	a81bb521	4cd9ab100d971208	06-14-2019 13:62:38	2021-01-11	Nintendo Switch	8d0d	120.26	website	direct	desktop	IE
7365	7c1b4739	60d6f8dd5dc53016	12-31-2020 01:54:34	2020-06-15	JBL Quantum 100 Gaming Headset	ab0f	21.46	website	direct	desktop	ES
9719	1db9d78a	7ed14bf67eb15182	12-31-2020 01:54:33	2021-01-03	Sony PlayStation 5 Bundle	54ed	1800	website	direct	desktop	US
10013	81d147e3	829839cc52155359	12-31-2020 01:54:44	2021-11-16	JBL Quantum 100 Gaming Headset	4db1	14.29	mobile app	direct	desktop	QA
10892	6dcd78e9	8d6795b3b1447377	12-31-2020 01:54:40	2020-02-20	27in 4K gaming monitor	891b	408	website	direct	mobile	US
11855	2fa9f33d	99d824517da22388	06-14-2019 13:62:39	2019-04-11	JBL Quantum 100 Gaming Headset	ab0f	21.19	website	direct	mobile	JP
14578	43bbad06	ba7f92bd22730500	12-31-2020 01:54:39	2019-03-31	JBL Quantum 100 Gaming Headset	ab0f	24	website	direct	desktop	US
16165	b313cea5	c9e0aea0d9a75871	06-14-2019 13:62:34	2019-05-18	JBL Quantum 100 Gaming Headset	ab0f	19.2	website	direct	desktop	US
20727	67f8050b	14de38506b644875	06-14-2019 13:62:41	2019-01-17	JBL Quantum 100 Gaming Headset	ab0f	25.69	website	direct	desktop	GB

The next issue is having two different spellings for the “assumed” same product in the product_name column:

7835	cff7e8e0	670e3996e5c87868		2020-03-03	2020-03-04	27in 4K gaming monitor
7845	bcda0957	672a556d21579949		2020-11-28	2020-12-01	27in 4K gaming monitor
7850	1a43454e	6732ce49d5413679		2020-04-22	2020-04-25	27in 4K gaming monitor
7851	9a019c6c	673347e7fd865100		2020-05-12	2020-05-13	27inches 4k gaming monitor
7856	38af6cfe	6748dc47b1725959		2020-12-09	2020-12-10	27in 4K gaming monitor
7858	39d4ac00	674ea9a747526415		2020-08-19	2020-08-21	27in 4K gaming monitor
7863	5f137a4a	675d2dd9aa241427		2020-11-18	2020-11-19	27in 4K gaming monitor

Looking at the count of both items, it seems that “27in 4K gaming monitor” is the correct spelling and the “27inches 4K gaming monitor” is incorrect.

After looking through the first table, here is our issue log:

Table	Column	Issue	Row Count	Solvable?	Resolution
orders	purchase_ts	inconsistent date formats	11		
orders	product_name	incorrect spelling of "27in 4K gaming monitor"	61		
orders	USD_price	Price of the item is "0"	29		
orders	marketing_channel	Cell is Blank	83		
orders	account_creation_method	Cell is Blank (same rows as blank marketing channel??)	83		
orders	country_code	Cell is Blank	37		

Next, we have our region table. This table has some country codes with missing or illogical values:

COUNTRY	REGION
IE	
LB	
MH	X.x
PG	X.x

Bringing together all the issues, we now have an improved issue log:

Table	Column	Issue	Row Count	Solvable?	Resolution
orders	purchase_ts	Inconsistent date formats (and one missing date)	11	Y-	Dates reformatted, missing date is removed.
orders	product_name	Incorrect spelling of "27in 4K gaming monitor"	61	Y	Spelling fixed
orders	USD_price	Price of the item is "0" or missing values	34	N	Ignored (can be handled differently based on business logic)
orders	marketing_channel	Missing values	83	Y	Blank cells assigned "unknown"
orders	account_creation_method	Missing values (same rows as blank marketing)	83	Y	Blank cells assigned "unknown"
orders	country_code	Missing values	37	N	No other region identifying information exists, thus not solveable.
region	region	Missing and nonsensical regions	4	Y	Country code can be looked up to match to regions.

Next, we need to identify the solvable issues. The inconsistent date formats can be corrected by reforming the data into the correct format, and the single missing date can be removed or ignored. The incorrect spelling issue can also be fixed similarly. The prices of items being zero is not an issue that can be fixed (unless the stakeholder allows us to use an averaged column value for them, or other such fixes). The marketing channel column and the account creation method column contain a datatype "unknown", which will be assigned to the columns' missing values. Missing country codes cannot be solved because there does not exist any additional identifiable geographic data to justify assumptions. Missing and nonsensical regions in the regions table can be fixed with a simple Google search, identifying the country codes (in an ideal situation, I would meet with the stakeholders to discuss this).

Now that we have completed our initial pass of the data, we can proceed to address the initial issues.

Each table is copied and renamed to "table_cleaned," and problematic columns are copied within the new table and renamed as "column_cleaned". This way, we ensure that the initial dataset remains unchanged, allowing us to visually compare the changes between the raw and cleaned columns.

Purchase_ts was intended to be a date column; however, this column had cells formatted as date AND text. It would be simple to fix this using Excel's format cells feature; however, I invited myself a challenge and improvised a formula to fix it:

```
=IF(ISNUMBER(C2), TEXT(C2, "yyyy-mm-dd"), TEXT(DATE(RIGHT(LEFT(C2,10),4), LEFT(C2,2), MID(C2,4,2)), "yyyy-mm-dd"))
```

The data gathered was based on an American company's format, resulting in value errors when using Excel's DATE function on Excel's Canadian version. This implementation checks if the cell is in date format; if yes, then it simply formats it using TEXT. In cases where it's not in date format, it extracts the individual date parts manually and reassembles them. The single unknown date value is blank.

Fixing the spelling mistake:

```
=IF(F2="27inches 4K gaming monitor", "27in 4K gaming monitor", F2)
```

Casting blanks as "unknowns":

```
=IF(ISBLANK(M2), "unknown", M2)
```

Lastly, the missing regions in the Region table are filled in. The following are snippets of the fixes in both tables (note the cleaned columns):

MARKETING_CHANNEL	MARKETING_CHANNEL_CLEANED	ACCOUNT_CREATION_METHOD	ACCOUNT_CREATION_METHOD_CLEANED
	unknown		unknown
	unknown		unknown
	unknown		unknown
	unknown		unknown

PRODUCT_NAME	PRODUCT_NAME_CLEANED
27inches 4k gaming monitor	27in 4K gaming monitor
27inches 4k gaming monitor	27in 4K gaming monitor

PURCHASE_TS	PURCHASE_TS_CLEANED
12-31-2020 01:54:35	2020-12-31
06-14-2019 13:62:38	2019-06-14
12-31-2020 01:54:34	2020-12-31
12-31-2020 01:54:33	2020-12-31

COUNTRY_	REGION	REGION_CLEANED
IE		EMEA
LB		EMEA
MH	X.x	APAC
PG	X.x	APAC

Now that we have our initial fixes, we can move on to utilizing pivot tables for deeper data cleaning. We will create a pivot table to find duplicate orders in our unique identity column: user_id.

1	Row Labels	Count of USER_ID
142	fecc9d6766231653	2
143	ee8e7c4961170394	2
144	fef061c2a1499684	2
145	eea8dd9fb5b67900	2
146	eed571de3da20556	2
147	Grand Total	290

From this, we know that there are 145 orders with duplicate user_ids. Examining the outlined order_ids from the row labels, we can see that they are indeed duplicate orders:

USER_ID	ORDER_ID	PURCHASE_TS	PURCHASE_TS_CLEANED	SHIP_TS	PRODUCT_NAME	PRODUCT_NAME_CLEANED	PRODUCT_ID	USD_PRICE	PURCHASE_PLATFORM
a0895503	eea8dd9fb5b67900	2021-02-17	2021-02-17	2021-02-19	Nintendo Switch	Nintendo Switch	8d0d	168	website
9628559	eea8dd9fb5b67900	2021-02-17	2021-02-17	2021-02-19	Nintendo Switch	Nintendo Switch	8d0d	168	website

Since our table has 20,000 rows, we can safely ignore this issue. In a real business context, the orders will be verified to be duplicates, and the handling process can be discussed with the stakeholders.

Next, we will evaluate the data using business logic. These sanity checks require domain knowledge and can span across tables. Looking at our table, we know that our data contains multiple date columns, purchase_ts and ship_ts. Understanding this, we know that an order can only be shipped after it has been purchased; as such, the value of purchase_ts must be lower than or equal to ship_ts (depending on business rules, we can enforce the value to be lower only). To do this check, we will create a new column called date check, where we will compare the date values:

	A	B	C	D	E	P
1	USER_ID	ORDER_ID	PURCHASE_TS	PURCHASE_TS_CLEANED	SHIP_TS	DATE_CHECK
2	2c06175e	0001328c3c220830	2020-12-24	2020-12-24	2020-12-13	After
3	ee8e5bc2	0002af7a5c6100772	2020-10-01	2020-10-01	2020-09-21	After
4	9eb4efe0	0002b8350e167074	2020-04-21	2020-04-21	2020-02-16	After
5	cac7cbaf	0006d06b98385729	2020-04-07	2020-04-07	2020-04-04	After
6	6b0230bc	00097279a2f46150	2020-11-24	2020-11-24	2020-08-02	After
7	046e7528	000f169e17f4941	2020-08-20	2020-08-20	2020-08-09	After
8	94dfec76	000fdd9bfe463101	2020-05-11	2020-05-11	2020-03-05	After

This check results in the largest magnitude issue so far, with 2000 rows where the purchase_ts is after the ship_ts. We can add these issues to our log:

A	B	C	D	E	F	G
Table	Column	Issue	Row Count	Magnitdue	Solvable?	Resolution
orders	purchase_ts	Inconsistent date formats (and one missing date)	11	0.05%	Y-	Dates reformatted, missing date is removed.
orders	product_name	Incorrect spelling of "27in 4K gaming monitor"	61	0.28%	Y	Spelling fixed
orders	USD_price	Price of the item is "0" or missing values	34	0.16%	N	Ignored (can be handled differently based on business logic)
orders	marketing_channel	Missing values	83	0.38%	Y	Blank cells assigned "unknown"
orders	account_creation_method	Missing values (same rows as blank marketing channel??)	83	0.38%	Y	Blank cells assigned "unknown"
orders	country_code	Missing values	37	0.17%	N	No other region identifying information exists, thus not solveable.
region	region	Missing and nonsensical regions	4	0.02%	Y	Country code can be looked up to match to regions.
orders	all	Duplicates (multiple user_ids reference same order_ids).	145	0.66%	N	Ignored (can be handled differently based on business logic)
orders	ship_ts and purchase_ts	Purchase timestamp after the ship timestamp	2000	9.15%	N	Ignored (can be handled differently based on business logic)

We can see that the magnitude of these issues is all under 10%, making it possible to generate good insights. We will proceed to create new columns within the data to help us calculate key performance indicators outlined by the stakeholders. We can choose to do this here or in MS Power BI. I will decide to do this in Power BI. Finally, the data cleaning is completed, and the final table is as follows:

A	B	C	D	E	F	G	H	I	J	K	L	M	
1	USER_ID	ORDER_ID	PURCHASE_TS	SHIP_TS	PRODUCT_NAME	PRODUCT_NAME	PRODUCT_ID	USD_PRICE	PURCHASE_PLATFORM	MARKETING_CHANNEL	ACCOUNT_CREATION_METHOD	COUNTRY_CODE	DATE_CHECK
2	2c06175e	0001328c3c220830	2020-12-24	2020-12-24	2020-12-13 Nintendo Switch	Nintendo Switch	e682	158	website	affiliate	unknown	US	After
3	ee8e5bc2	0002af7a5c6100772	2020-10-01	2020-09-21	Nintendo Switch	Nintendo Switch	e682	160.61	website	direct	desktop	DE	After
4	9eb4efe0	0002b8350e167074	2020-04-21	2020-02-16	Nintendo Switch	Nintendo Switch	8d0d	151.2	website	direct	desktop	US	After
5	cac7cbaf	0006d06b98385729	2020-04-07	2020-04-04	Sony PlayStation 5 Bundle	Sony PlayStation 5 E 54ed		1132.82	website	direct	desktop	AU	After

Next, we will derive insights from this data using Power BI (explored in the README.md).

GameZone Sales Revenue Analysis (2018-2021)

Year	Region	Month
All	All	All

Sales Analysis

9.65%

YoY Growth

22K

Sales Amount

\$6.15M

Total Revenue

3.42

TTS (DAYS)

Growth Analysis

0.05%

MoM Growth

Detailed Breakdown

\$281.34

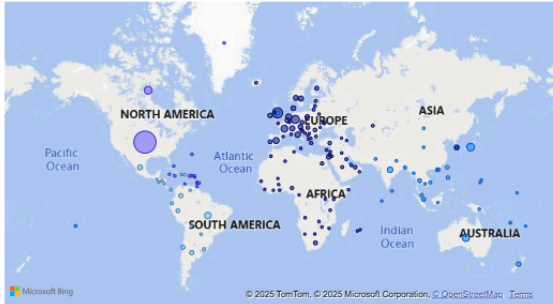
AOV

Nintendo Switch

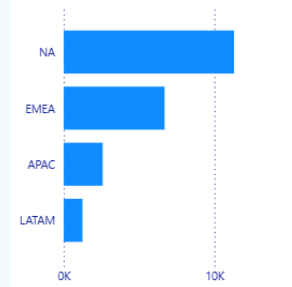
Best Selling Product

Revenue Density by Location

Region ● APAC ● EMEA ● LATAM ● NA



Sales Amount by Region



Revenue by Region

